

MINIMUM HARDWARE SPECIFICATIONS FOR CLIMATE ATLAS PROCESSING WORKFLOWS

The following hardware specifications represent the minimum recommended requirements for the execution of the Climate Atlas processing methodologies. The workflows include the execution of multiple Jupyter notebooks dedicated to climate data download, preprocessing of very high resolution geospatial variables and reanalysis datasets, statistical downscaling procedures, inverse distance weighting (IDW) interpolation based on meteorological station observations, and the generation of integrated NetCDF outputs and PNG visualization products.

The computational requirements depend on the specific Atlas module (Solar, Wind, or Hydropower), the spatial extent of the target country, and the amount of climate data processed simultaneously. In particular, Solar and Wind Atlas workflows require substantial memory and storage resources due to the manipulation of high resolution raster datasets and multi-variable climate simulations.

The table below summarizes the minimum recommended hardware specifications for operational execution of the workflows

Component	Atlas	Minimum Specification	Notes
CPU	All	Minimum 4 CPU cores	Strong single-core and multi-core performance recommended for preprocessing, geospatial operations, and parallel workflows
RAM	Solar, Wind	16 GB per 200,000 km ²	Required for high-resolution downscaling and large raster processing
RAM	Hydropower	16 GB	Hydrological aggregation on sub-basins generally avoids spatial splitting during processing
Swap	All	Half of RAM size	Recommended to avoid memory crashes during large computations
Disk storage	Solar	100 GB	Estimate valid for one selected climate model, scenario, and period
Disk storage	Wind	200 GB	Includes storage for u/v wind component downscaling and intermediate outputs
Disk storage	Hydropower	25 GB	Outputs mainly stored as GeoJSON and CSV; HydroSHEDS data reused across case studies

Setting Up the Environment for Executing Anaconda

Visit the Anaconda Downloads page (<https://www.anaconda.com/products/distribution>). Select the appropriate installer for your operating system:

For Windows: Choose the 64-bit installer (.exe file).

For Ubuntu: Choose the Linux installer (.sh file).

Installation on Windows

Run the downloaded .exe installer and follow the installation wizard:

1. Click 'Next' to proceed through the initial steps.
2. Accept the license agreement.
3. Select an installation folder (default is recommended).
4. Choose to 'Add Anaconda to my PATH environment variable' (optional, but not recommended).
5. Install Anaconda.

Verify Installation:

1. Open the Anaconda Navigator from the Start Menu.
2. Alternatively, open the Command Prompt and type:

```
conda --version
```

```
conda --version
```

This should return the installed version of Conda.

Installation on Ubuntu

Open a terminal and navigate to the folder containing the downloaded .sh file. Execute the following:

```
chmod +x Anaconda3*.sh
```

```
./Anaconda3*.sh
```

Follow the prompts:

1. Read and agree to the license terms.
2. Choose the installation directory (default is recommended).
3. Confirm adding Anaconda to your PATH during installation.

Activate and Verify Installation:

```
source ~/anaconda3/bin/activate
conda --version
```

Post-Installation Setup on Windows

Open Anaconda Navigator and click on Environments. In the Channels tab, add a channel and paste this url: <https://conda.anaconda.org/conda-forge/>

Click on Import and load the environment yml, e.g. Geo_requirements.yml.

Post-Installation Setup on Ubuntu

Update Conda:

```
conda update -n base -c defaults conda
```

Create a new environment for the first iteration and, with different yml, for the following ones:

```
conda create -n geo_requirements -f Geo_requirements.yml
conda activate geo_requirements
```

Ubuntu: Use the terminal to activate the environment and run commands. Optionally, install Anaconda Navigator GUI:

```
conda install anaconda-navigator
anaconda-navigator
```

Opening JupyterLab

Launch JupyterLab and import the selected notebook (e.g. Orography_processing.ipynb) with the environment created in the step before.

N.B. Download and place the package data in the same folder of the running Jupyter Notebook.